

Theorem 5.3.1 For all integers

$$n, 2^n - 1$$

is divisible by 3.

Proof Let the property

$$P(n)$$

be the sentence

$$2^n - 1 \text{ is divisible by } 3.$$

$$2^n - 1 \text{ is divisible by } 3.$$

Show that

$$P(0)$$

is true: To establish

$$P(0)$$

, we must show that

$$2^{2^0} - 1$$

is divisible 3.

But

$$2^{2^0} - 1 = 2^0 - 1 = 1 - 1 = 0$$

and 0 is divisible by 3 because

$$0 = 3 * 0$$

. Hence

$$P(0)$$

is true.

Show that for all integers k , if $P(k)$ is true then $P(k + 1)$ is also true.